OBSTACLE AVOIDANCE ROBOT COMPONENTS LIST

S.No.	Component	Description	Qty
1	Arduino Uno R3	ATmega328P-based microcontroller board responsible for sensor interfacing, motor control, and execution of the obstacle avoidance algorithm.	1
2	L293D Motor Driver Shield	Dual H-Bridge motor driver used to control the direction and speed of the DC gear motors through PWM signals from the Arduino.	1
3	HC-SR04 Ultrasonic Sensor	Ultrasonic ranging sensor with a 2–400 cm sensing range, used to detect and measure the distance to nearby obstacles.	1
4	SG90 Micro Servo Motor	180° micro servo motor used to rotate the ultrasonic sensor for scanning the surrounding environment.	1
5	DC Gear Motor	12 V, 100 RPM centre-shaft DC geared motors used to provide propulsion for the robot.	2
6	HM-10 BLE 4.0 Bluetooth Module	Enables wireless serial communication between the robot and a Bluetooth-enabled mobile device for manual operation.	1
7	LM2596 Buck Converter	DC-DC buck converter used to regulate the battery voltage to a stable 5 V supply for the control electronics.	1
8	18650 Li-ion Battery	Rechargeable 3.7 V, 2200 mAh lithium-ion cells used as the primary power source for the robot.	3
9	3-Slot 18650 Battery Holder	Holds three 18650 lithium-ion batteries securely while providing electrical connections to the power circuit.	1
10	SPDT 3-Pin Toggle Switch	Three-pin Single Pole Double Throw (SPDT) switch used to connect or disconnect the main power supply.	1
11	MDF Board	Serves as the structural chassis for mounting electronic and mechanical components.	1
12	34 mm Rubber Wheel	Mounted on the DC gear motors to convert rotational motion into linear movement of the robot.	2
13	1-inch Swivel Caster Wheel	Provides passive support and enables smooth multidirectional movement while maintaining chassis stability.	1
14	Perfboard	General-purpose prototyping board used for assembling and soldering auxiliary electronic circuits.	1
15	Jumper Wires	Used for temporary electrical interconnections during circuit assembly, testing, and debugging.	As Required
16	Single Strand Copper Wires	Used for permanent soldered electrical connections between electronic components.	As Required
17	Screws, Nuts & Spacers	Mechanical fasteners used to securely mount the electronic and mechanical components onto the chassis.	As Required

OPTIONAL COMPONENTS FOR ADDING CLEANING FUNCTIONALITY

S.No.	Component	Description	Qty
1	BO Motor with Mop Attachment	DC motor with mop attachment used to perform floor cleaning operation.	1
2	18650 Li-ion Battery	Dedicated 3.7 V, 2200 mAh lithium-ion battery to power the cleaning module.	1

3	Single-Cell 18650 Battery Holder	Holder used to securely hold the single 18650 battery and provide electrical connections.	1
4	SPDT 3-Pin Toggle Switch	Switch used to independently control the power supply to the cleaning motor.	1

Note: The optional components are used only when the cleaning functionality is enabled. They have a separate power source and switch to operate independently of the main robot.